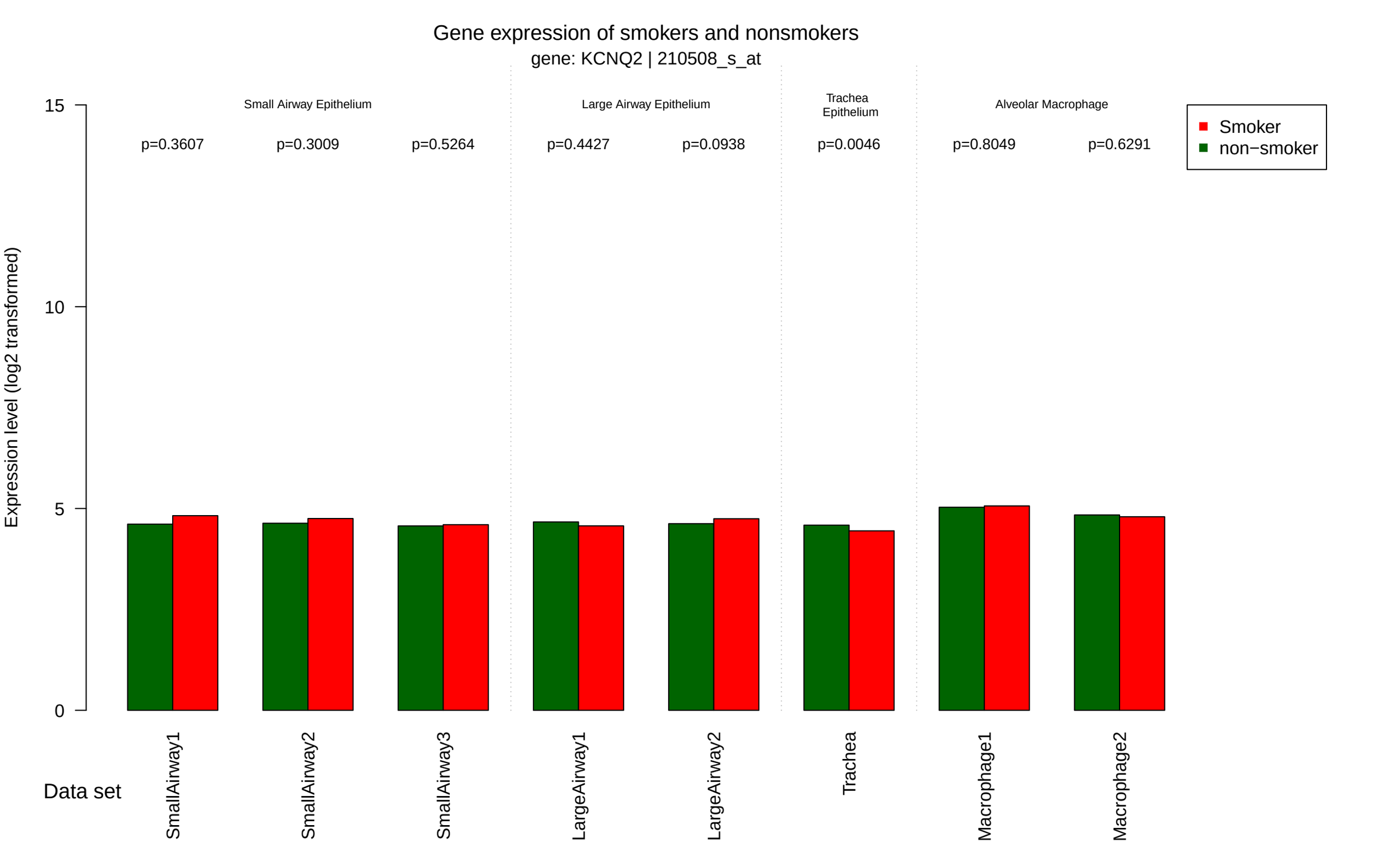
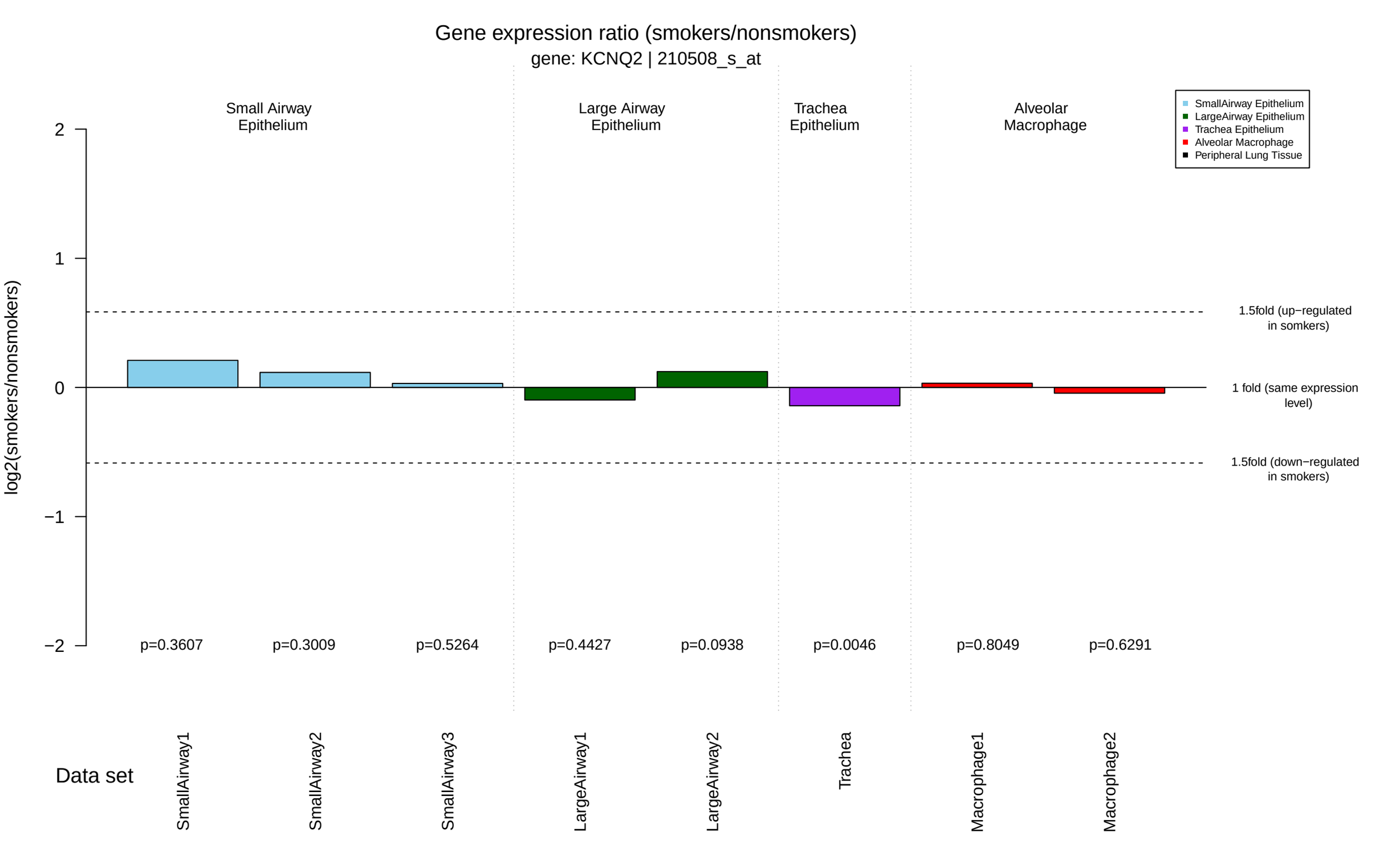
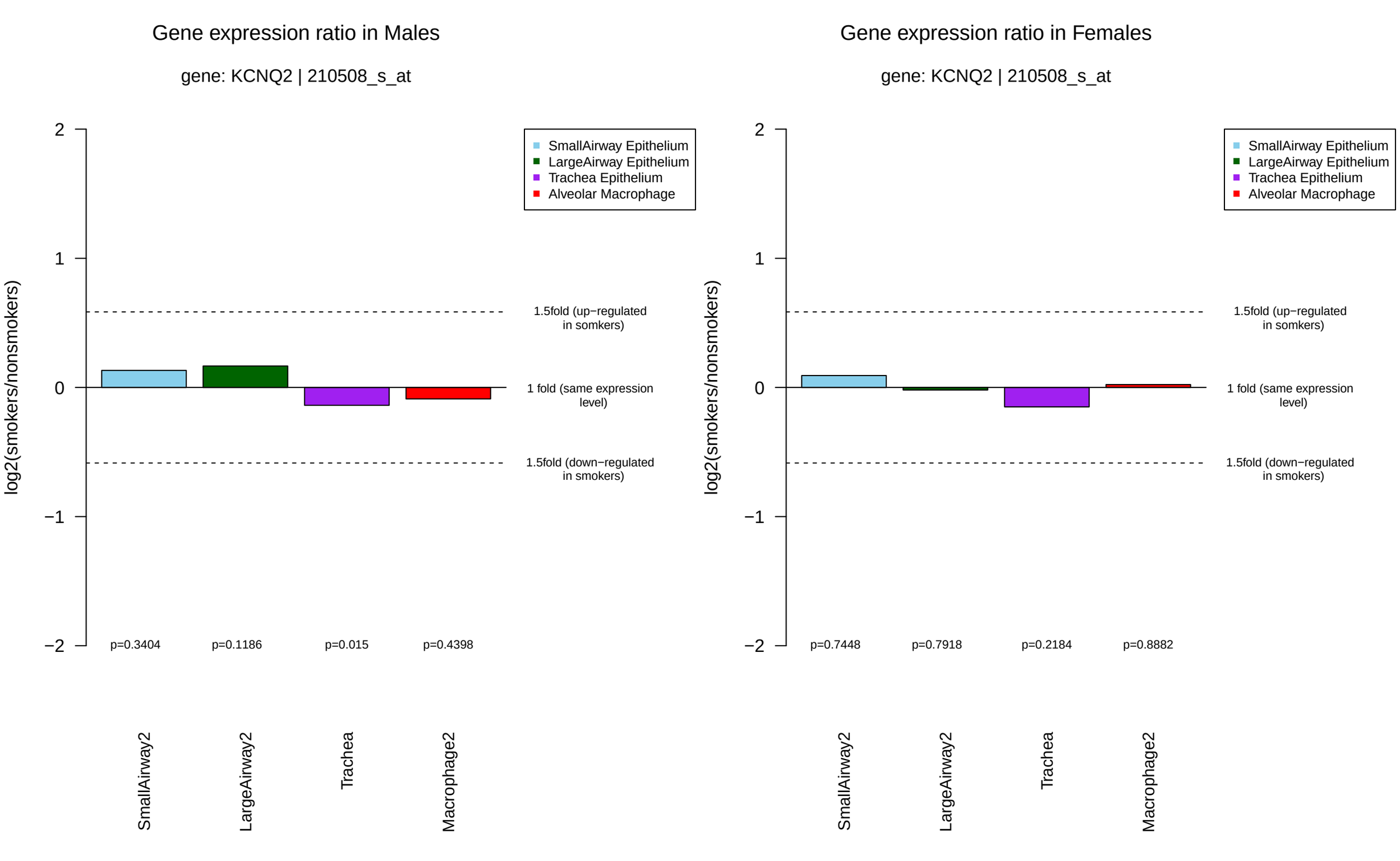
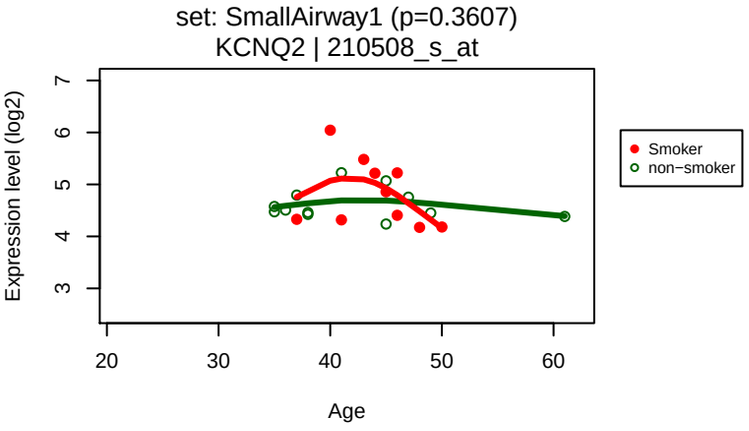
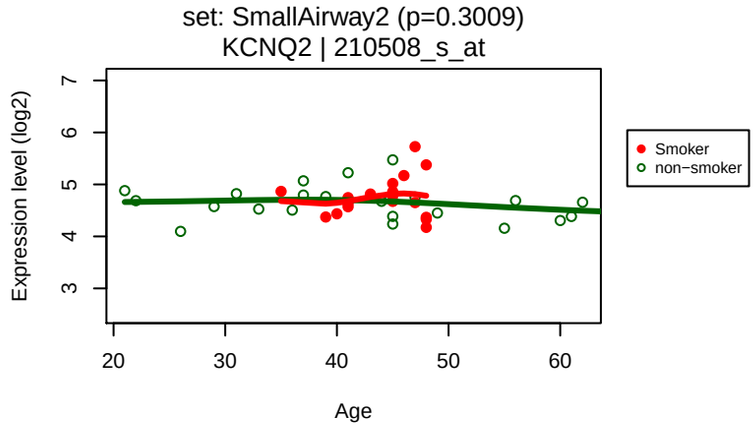
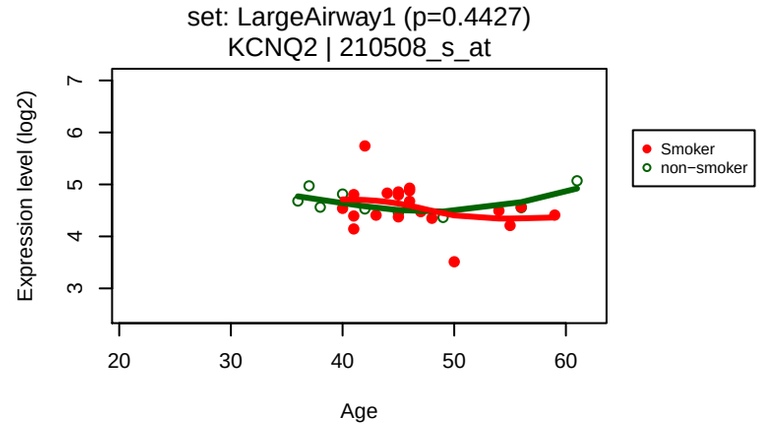

Gene expression of smokers and nonsmokers
gene: KCNQ2 | 210508_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3607
p=0.3009
p=0.5264
p=0.4427
p=0.0938
p=0.0046
p=0.8049
p=0.6291
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

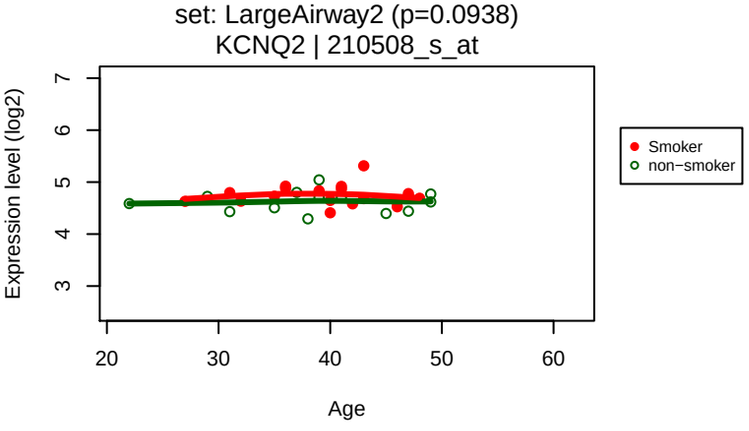

Gene expression ratio (smokers/nonsmokers)
gene: KCNQ2 | 210508_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3607
p=0.3009
p=0.5264
p=0.4427
p=0.0938
p=0.0046
p=0.8049
p=0.6291
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

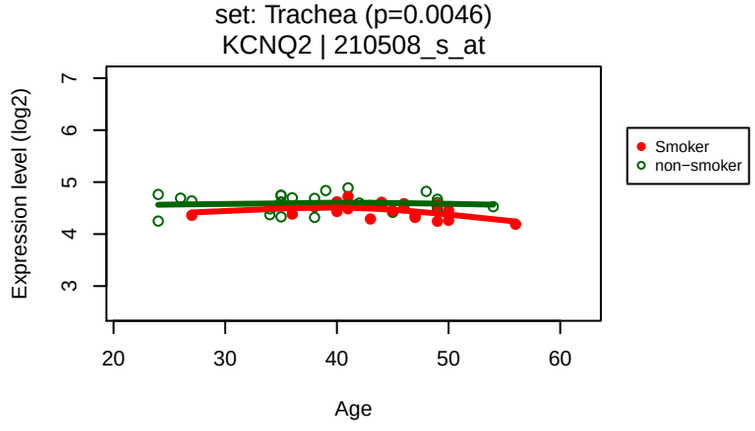

Gene expression ratio in Males
gene: KCNQ2 | 210508_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3404
p=0.1186
p=0.015
p=0.4398
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNQ2 | 210508_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7448
p=0.7918
p=0.2184
p=0.8882
SmallAirway2
LargeAirway2
Trachea
Macrophage2

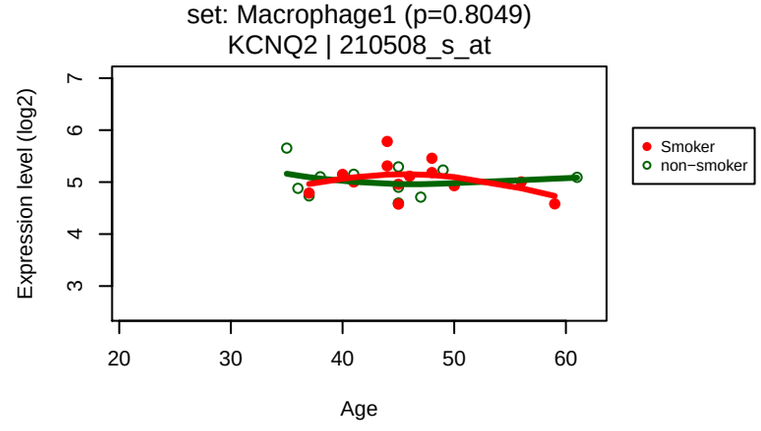

set: SmallAirway1 (p=0.3607)
KCNQ2 | 210508_s_at
Expression level (log2)
Age
Smoker
non−smoker

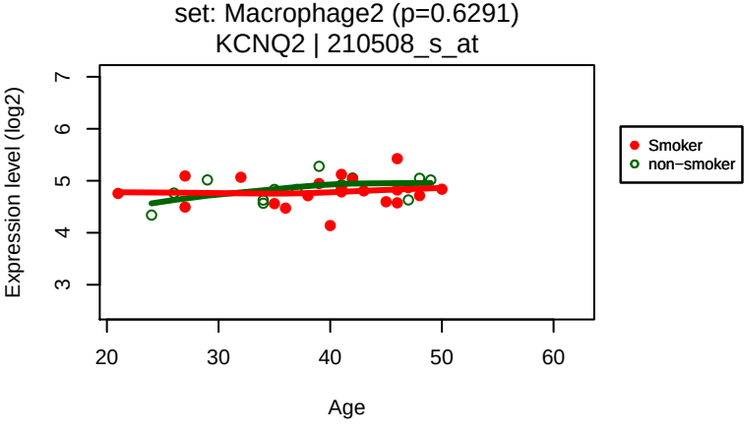

set: SmallAirway2 (p=0.3009)
KCNQ2 | 210508_s_at
Expression level (log2)
Age
Smoker
non−smoker

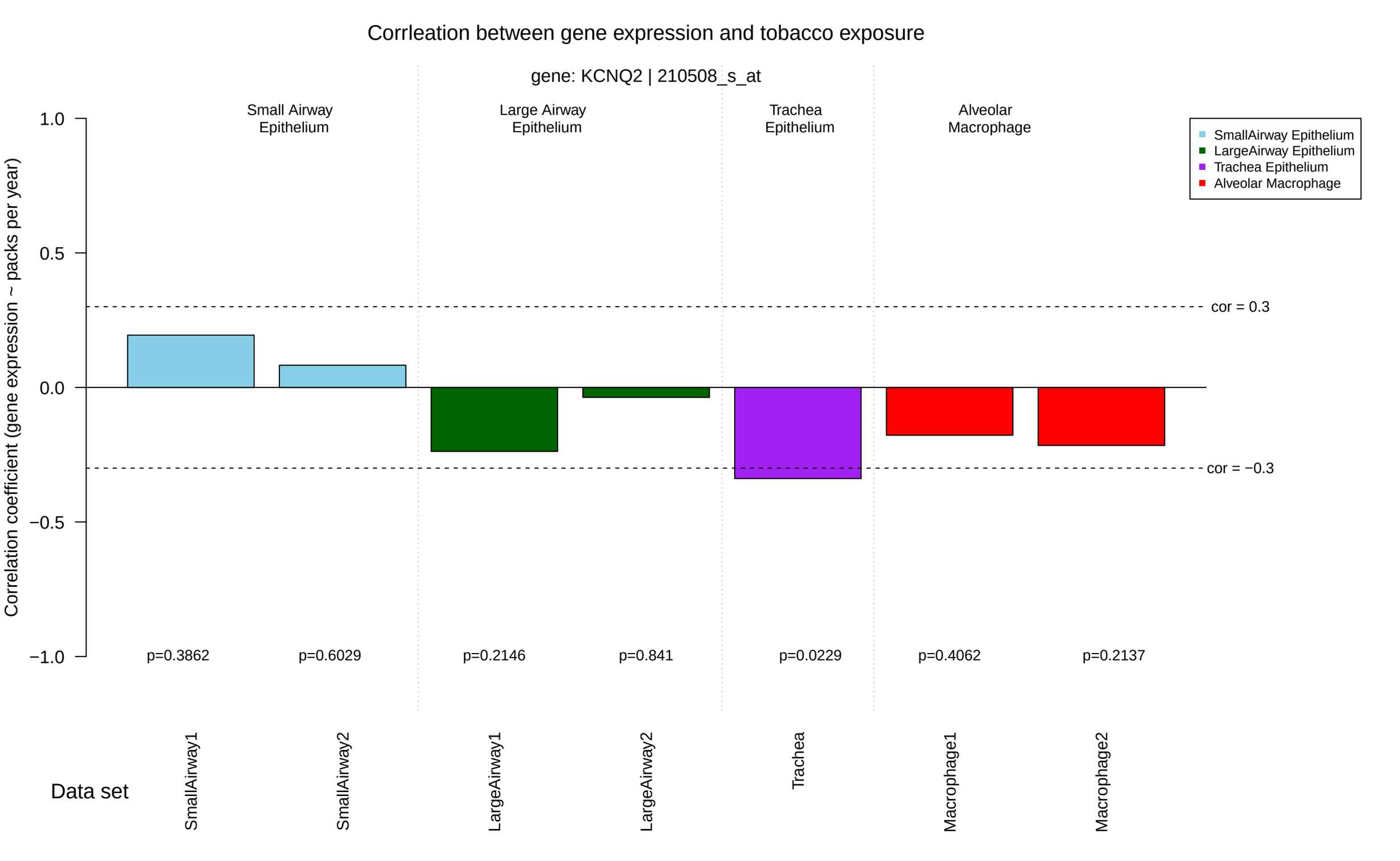

set: LargeAirway1 (p=0.4427)
KCNQ2 | 210508_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0938)
KCNQ2 | 210508_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0046)
KCNQ2 | 210508_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8049)
KCNQ2 | 210508_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6291)
KCNQ2 | 210508_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNQ2 | 210508_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3862
p=0.6029
p=0.2146
p=0.841
p=0.0229
p=0.4062
p=0.2137
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2